

5.2.44

EE25BTECH11042 - Nipun Dasari

Question:

Solve the following system of rational equations

$$\frac{10}{x+y} + \frac{2}{x-y} = 4 \quad (0.1)$$

$$\frac{15}{x+y} - \frac{5}{x-y} = -2 \quad (0.2)$$

Solution:

We convert the system of rational equations from (0.1) and (0.2) into the standard quadratic form for conic sections. For the first equation (0.1), we multiply by $(x+y)(x-y)$:

$$\begin{aligned} 10(x-y) + 2(x+y) &= 4(x^2 - y^2) \\ 12x - 8y &= 4x^2 - 4y^2 \\ \implies x^2 - y^2 - 3x + 2y &= 0 \end{aligned} \quad (0.3)$$

For the second equation (0.2), we do the same:

$$\begin{aligned} 15(x-y) - 5(x+y) &= -2(x^2 - y^2) \\ 10x - 20y &= -2x^2 + 2y^2 \\ \implies x^2 - y^2 + 5x - 10y &= 0 \end{aligned} \quad (0.4)$$

The intersection points lie on a degenerate conic formed from a linear combination $H_1 + \mu H_2 = 0$. We must find the value of μ that makes the determinant of the combined conic's matrix equal to zero. The general matrix for the conic pencil $H_1 + \mu H_2$ is:

$$C(\mu) = \begin{pmatrix} 1+\mu & 0 & \frac{5\mu-3}{2} \\ 0 & -(1+\mu) & 1-5\mu \\ \frac{5\mu-3}{2} & 1-5\mu & 0 \end{pmatrix} \quad (0.5)$$

We find the determinant by reducing the matrix to row echelon form.

$$C(\mu) \xrightarrow{R_3 \rightarrow R_3 - \frac{5\mu-3}{2(1+\mu)} R_1} \begin{pmatrix} 1+\mu & 0 & \frac{5\mu-3}{2} \\ 0 & -(1+\mu) & 1-5\mu \\ 0 & 1-5\mu & -\frac{(5\mu-3)^2}{4(1+\mu)} \end{pmatrix} \quad (0.6)$$

$$\xrightarrow{R_3 \rightarrow R_3 + \frac{1-5\mu}{1+\mu} R_2} \begin{pmatrix} 1+\mu & 0 & \frac{5\mu-3}{2} \\ 0 & -(1+\mu) & 1-5\mu \\ 0 & 0 & \frac{5(15\mu^2-2\mu-1)}{4(1+\mu)} \end{pmatrix} \quad (0.7)$$

The determinant is the product of the diagonal entries. We set it to zero:

$$\begin{aligned}\det(C(\mu)) &= (1 + \mu) \cdot (-(1 + \mu)) \cdot \left(\frac{5(15\mu^2 - 2\mu - 1)}{4(1 + \mu)} \right) = 0 \\ \implies -\frac{5(1 + \mu)(15\mu^2 - 2\mu - 1)}{4} &= 0\end{aligned}\quad (0.8)$$

This equation is satisfied if either $(1 + \mu) = 0$ or $(15\mu^2 - 2\mu - 1) = 0$. The simplest root is obtained from the first factor, which gives $\mu = -1$. The equation for this degenerate conic is found by subtracting (0.4) from (0.3):

$$\begin{aligned}(x^2 - y^2 - 3x + 2y) - (x^2 - y^2 + 5x - 10y) &= 0 \\ -8x + 12y &= 0 \\ \implies y &= \frac{2}{3}x\end{aligned}\quad (0.9)$$

We find the intersection of the common chord from (0.9) with H_1 using the formula $\mathbf{x}_i = \mathbf{h} + \kappa_i \mathbf{m}$. The line and conic parameters from H_1 are: $\mathbf{h} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$, $\mathbf{m} = \begin{pmatrix} 3 \\ 2 \end{pmatrix}$, $\mathbf{V} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$, $\mathbf{u} = \begin{pmatrix} -3/2 \\ 1 \end{pmatrix}$, and $f = 0$.

Next, we compute the components for the parameter κ_i :

$$\mathbf{m}^\top \mathbf{V} \mathbf{m} = 5 \quad (0.10)$$

$$g(\mathbf{h}) = 0 \quad (0.11)$$

$$\mathbf{m}^\top (\mathbf{V} \mathbf{h} + \mathbf{u}) = -\frac{5}{2} \quad (0.12)$$

We substitute these into the formula for κ_i :

$$\begin{aligned}\kappa_i &= \frac{1}{\mathbf{m}^\top \mathbf{V} \mathbf{m}} \left(-\mathbf{m}^\top (\mathbf{V} \mathbf{h} + \mathbf{u}) \pm \sqrt{[\mathbf{m}^\top (\mathbf{V} \mathbf{h} + \mathbf{u})]^2 - g(\mathbf{h})(\mathbf{m}^\top \mathbf{V} \mathbf{m})} \right) \\ &= \frac{1}{5} \left(-(-\frac{5}{2}) \pm \sqrt{(-\frac{5}{2})^2 - (0)(5)} \right) = \frac{1}{5} \left(\frac{5}{2} \pm \frac{5}{2} \right)\end{aligned}\quad (0.13)$$

This gives two values: $\kappa_1 = 1$ and $\kappa_2 = 0$.

Finally, we find the intersection points using $\mathbf{x}_i = \mathbf{h} + \kappa_i \mathbf{m}$:

$$\mathbf{x}_1 = \begin{pmatrix} 0 \\ 0 \end{pmatrix} + 1 \begin{pmatrix} 3 \\ 2 \end{pmatrix} = \begin{pmatrix} 3 \\ 2 \end{pmatrix} \quad (0.14)$$

$$\mathbf{x}_2 = \begin{pmatrix} 0 \\ 0 \end{pmatrix} + 0 \begin{pmatrix} 3 \\ 2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \quad (0.15)$$

The intersection points are (3, 2) from (0.14) and (0, 0) from (0.15). The point (0, 0) is extraneous because it causes division by zero in the original equations (0.1) and (0.2). Thus, the only valid solution is $x = 3, y = 2$.